

ERP Transformation Program

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Strategic Recommendations for Alpha Manufacturing Group

To:AMG Executive Team

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Subject:ERP Replacement — Approach, Scope & Data Strategy

AMG has an 80% margin erosion problem caused by a broken handoff between Engineering, Manufacturing, and Procurement. The overnight BOM sync fails. The factory floor builds from stale parts lists. And nobody sees it coming until the P&L lands months later. This memo lays out a 12-month plan to fix that — without breaking the business along the way. The strategy is organized around three dimensions: program structure and phasing, scope discipline, and data continuity.

1. Overall Approach

Key Objectives

The Board's mandate translates into four concrete objectives that define what "success" looks like at the end of this program:

1. Stop losing money blind on custom orders — Right now 80% of custom jobs lose margin and nobody sees it coming. By go-live, leadership must be able to see, in real time, whether a job in progress is on track to make or lose money.

2. Kill the overnight BOM sync failure problem — Engineering and the factory floor must be looking at the same parts list, always — not a list that might be a day stale because a nightly sync broke.

3. Don't break the business while fixing it — Zero missed customer shipments. Zero failed month-end closes. Throughout the entire 12-month project.

4. Stop the Excel band-aid habit — Reduce dependency on shadow spreadsheets by giving people trustworthy data inside the ERP itself.

Phasing Approach — The Core Philosophy

AMG's biggest operational headache is the broken handoff between Engineering (designs the machine) and Manufacturing/Procurement (builds it). That overnight BOM sync failure is causing real delays and real lost money.

The Plumbing Rule: You don't rip out half the plumbing in your kitchen and leave the other half — that's how you flood the house. Engineering, Manufacturing, and Procurement are AMG's plumbing. They are connected and must move together as one go-live event. Everything else — HR, CRM, satellite sites — can be the guest bedroom: renovated later, phased gradually.

This leads to a hybrid phasing structure. The core (Engineering + Manufacturing/MRP + Procurement) goes live as one connected unit. Other modules and lower-risk sites follow in controlled waves after the core is stable.

Four-Phase Timeline

Phase	Timeline	Key Activities
Discovery & Design	Months 1–3	Map end-to-end Engineering→Manufacturing→Finance process; select cloud ERP; establish Scope Control Gate; begin data cleansing of legacy records
Build & Configure	Months 4–8	Configure core Finance + Mfg/MRP + Procurement as one connected suite; integrate Engineering BOM system; build margin visibility dashboards; run mock data migrations

Test & Rehearse	Months 9–10	Short parallel run — both old and new systems run side-by-side. Dry-run month-end close. UAT at all 5 locations. Validate BOM integration with real production data
Go-Live & Stabilize	Months 11–12	Core system cutover. Old system kept as read-only. 30-day hypercare. Satellite site rollout prep begins

Cutover Strategy — "Switching Banks"

A pure big-bang cutover (flip the switch, old system off, everyone on new Monday morning) is too risky here. AMG already doesn't fully trust their numbers — that's the whole reason 80% of orders lose money. Flipping to a brand new system with zero verification would compound the exact problem they're trying to fix.

The Switching Banks Rule: When you open a new bank account, you don't close the old one on the same day. You keep both open for a month, make sure paychecks and bills land correctly in the new one, then close the old one. Same logic here. A short parallel run (2–4 weeks, not months — we don't want permanent double-entry fatigue) around the core go-live lets us cross-check the new system's numbers against the old one before committing.

Top 5 Risks & Mitigations

#	Risk	Mitigation
1	BOM/engineering data doesn't migrate cleanly — Their engineering DB is already fragile. If messy data goes in, the new system inherits it.	Clean before you move. Audit the engineering database in Phase 1. Fix errors and de-duplicate before any data touches the new system. Don't migrate first and clean later.
2	Custom-order complexity gets underestimated — Engineer-to-order creates more variation than a standard-product ERP handles easily.	Test with real messy orders. Pick a handful of AMG's most complex past orders and run them through the new system before go-live. Prove it works on real complexity, not simplified test scenarios.
3	Scope creep from endless customizations — Every department wants the new system to replicate their old workarounds exactly.	Scope Control Gate (see separate section below). One small group, default answer is "defer." Customization requests must prove the business <i>cannot operate</i> without them.
4	Month-end close breaks during transition — Finance must close the books every single month, no matter what phase the project is in.	Never schedule cutover near close week. Build the project calendar around Finance's calendar. Dry-run close in Month 10. Keep legacy close as fallback for one full cycle.
5	Multi-entity consolidation gets tangled — 5 locations, multi-entity structure. If sites go live at different times, consolidated financial reporting can get messy.	Temporary manual reconciliation. During the mixed period (some sites old, some new), use a simple standard checklist to consolidate numbers across both systems by hand so leadership isn't flying blind.

Scope Control — The Gate That Keeps the Timeline Alive

Scope creep — not technical difficulty — is what kills ERP project timelines in the real world. Scope control needs its own explicit mechanism, not just a mention inside a risk section.

- **Who:** A small cross-functional gate-keeping group (VP Ops, CFO, Program Lead) — not a big committee. Small groups make faster, firmer calls.
- **The rule:** The default answer to any scope request is "**not now.**" Only requests that meet the "the business breaks without it" threshold get approved for Day-1. Everything else is logged for Phase 2.
- **What this protects:** The 12-month timeline. Every "small customization" that slips in quietly pushes the go-live date. This gate forces honest prioritization.
- **Connection to Prompt 2:** This gate only works if there's a clear Day-1 must-have list to measure against — which is exactly what Prompt 2 defines.

Decision Rule: If AMG cannot operate safely, invoice customers, or ship product without a proposed feature, it is Day-1 essential. If it would be merely inconvenient to defer, it *must* be deferred. "We've always done it this way" is not a valid reason.

2. Scope & Sequencing

Day-1 Must-Haves

The test for "must-have" is existential: *can AMG run its business without this on Day 1?* These four capabilities clear that bar:

- **Connected Engineering → Manufacturing → Procurement flow** — This is the central reason the project exists. The BOM/ECN sync failures are the #1 operational pain point. The ERP must receive engineering BOMs accurately and push them through MRP to procurement without the current nightly-sync fragility.
- **Core Finance** — General ledger, accounts payable, accounts receivable, and the ability to close the books monthly. A company cannot stop invoicing or paying suppliers.
- **Basic cost & margin visibility per order** — Even a simplified first version of "what is this job costing us as it's built." This directly addresses the 80% margin erosion problem. It does not need to be perfect on Day 1, but it must exist.
- **Inventory accuracy** — Manufacturing depends on knowing what parts are actually in stock. This must be live and correct from Day 1.

Deliberately Deferred

These are legitimate needs — just not needs that stop the business if they wait:

- **CRM integration** — Already unintegrated today; nothing gets worse by waiting.
- **HRIS/Payroll integration** — Already siloed and functioning. Not urgent enough to risk the core go-live.
- **Advanced analytics & predictive reporting** — Basic visibility is Day-1; the polished version comes once data is trustworthy.
- **Satellite site rollouts** — Lower-volume locations can follow in a second wave after core sites stabilize.

Migration Exclusions

Not everything from 20 years of history needs to move into the new system. The rule: **migrate what people need to keep working, archive what they might need to look up, and leave behind what nobody has touched in years.**

- **Archive, don't migrate:** Fully closed-out orders and service tickets older than 3 years — keep accessible in a read-only archive, not in the live system.
- **Clean out:** Dead/inactive vendor records, duplicate customer entries, obsolete part numbers. A new system with junk data defeats the purpose.
- **Validate before trusting:** Run migrated numbers side-by-side against the old system for a representative sample (e.g., last year's closed orders) before go-live.

Customization Discipline

Engineer-to-order manufacturers are notorious for wanting to customize everything. Resisting that is what keeps the 12-month timeline alive. Every customization request must pass through the Scope Control Gate and answer: *"If we don't build this, does the business actually break?"* If not, it is logged for Phase 2. The default is to adopt the ERP's standard workflows and adapt AMG's processes to match — not the reverse.

Customization Rule: Default to standard. Customize only when the business cannot operate without it. "The old system did it this way" is not a valid reason.

3. Data & Reporting Continuity

Historical Data Migration

The approach splits history into two tiers:

- **Active history** (open orders, current BOMs, active customers and vendors, recent financials) — migrates fully into the new system.
- **Archive history** (closed orders older than 3 years, inactive records) — moved to a searchable, read-only archive. Still accessible, but not cluttering the live system.

Prior to go-live, a **side-by-side validation** is run on a sample set (e.g., last 12 months of orders) to confirm no data corruption occurred during migration.

Reporting During Transition

Leadership cannot lose visibility into business performance during the migration. The strategy:

- **Single reporting source of truth** — Even if initially manually stitched from whichever system holds the live number for each KPI, this prevents confusion.
- **No reporting blackout** — During the parallel run (Months 9–10), reports are generated from both systems and compared to validate the new ERP's outputs.
- **Key KPIs to track through transition:** Contribution margin per order, on-time delivery rate, inventory turnover, BOM accuracy rate, and manufacturing schedule adherence.

Month-End Close Continuity

The month-end close is the most concrete risk in the entire project — it runs every month without exception.

- **Never cut over during close week.** This is a hard rule, not a guideline.
- **Conduct a full dry-run close** on the new system in Month 10 — before it becomes the official process — so Finance isn't learning the system for the first time on a day that counts.
- **Keep the old system's close process as a fallback** for one full cycle after go-live. If the new system's close hits a snag, Finance has a path to close the books on time.

4. Success Measures

The program's success should be measured against the Board's three mandate items with specific, observable criteria:

Mandate	Success Criteria	Measurement
1. Modern cloud ERP	Legacy system decommissioned; core business functions running on new ERP across all locations	By Month 12, >90% of transactions processed on new system; legacy in read-only
2. Data governance & rationalization	Single source of truth for BOM, inventory, and financial data; CRM integration underway	BOM sync failures reduced to <1%; no manual data re-entry between systems
3. Margin transparency	Reliable per-order contribution margin visible to operations and leadership in real time	80%+ of custom orders show tracked margin; variance reporting automated monthly

5. Anticipated Questions for Live Review

Based on the evaluation criteria (Structure, Prioritization, Judgment, Presentation), the following questions are likely during the solution defense:

- **"Why a phased approach instead of big bang?"** — Big bang introduces unacceptable risk for a company that cannot afford operational downtime. Phased allows fallback, builds confidence, and de-risks the BOM integration.
- **"What would cause you to delay go-live?"** — If the dry-run close reveals material data integrity issues, if BOM integration is not validated with real production data, or if user training completion rates fall below 85%.
- **"How do you prevent the project from becoming a custom development exercise?"** — The Scope Control Gate with its "default to defer" rule, the explicit customization decision framework, and the 12-month timeline as a non-negotiable constraint all reinforce standardization.
- **"What happens if the parallel run shows discrepancies?"** — Discrepancies are expected and valuable. They are logged, investigated, and resolved before cutover. The parallel run exists precisely to surface these issues in a safe environment.
- **"How do you ensure the 5-location rollout doesn't cascade into failure?"** — Core sites go first (highest volume, highest complexity); satellite sites follow in a second wave once the system is stable. Each location has a designated super-user trained before go-live.
- **"What is your data retention policy for the legacy system post-go-live?"** — Legacy system is kept in read-only mode for 6 months post-cutover, then decommissioned. Archived data remains accessible via a separate archive interface.

Executive Summary: AMG's ERP transformation is achievable within 12 months through disciplined phasing, a rigorous scope control mechanism, and a data strategy that prioritizes continuity. The recommendations above are designed to be defended not as theory, but as practical decisions grounded in AMG's specific operational reality — fragmented systems, BOM sync fragility, and the urgent need for margin visibility.

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Deliverable: Executive Memo · Supporting presentation available separately